

Applied Linear Statistical Models

Lecture 11 : Bayesian Linear Regression

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Academic Year 2025-2026

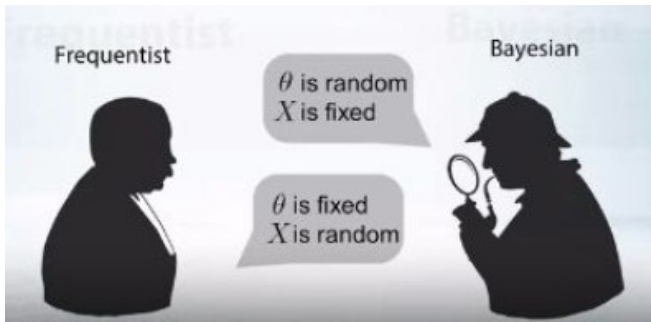
Outline

- 1 Review
- 2 Bayesian Linear Regression
- 3 Predictive Distribution
- 4 Regularization
- 5 Summary

Section 1

Review

Review



Frequentist: parameters are fixed unknown numbers.

Bayesian: parameters themselves are uncertain.

Review

Frequentist:

So far, learning a parametric model has amounted to somehow finding a parameter value $\hat{\theta}$ that best fits the training data.

- OLS
- MLE

OLS/MLE give one best estimate.

Review

The Bayesian approach (also called the *probabilistic approach*),

- Now, we still observe data, assumed to be randomly generated by some process associated with some unknown object.
- We have a **prior belief** about the object (e.g., parameters).
- Using the data, we want to **update that belief** and transform it into a **posterior** belief.
 - the learning amounts to find the **distribution** of parameter values θ conditioned on the observed training data D , that is, $p(\theta|D)$.
- Furthermore, with the Bayesian approach, **the prediction** is a **distribution** $p(y_*|x_*, D)$ instead of a single value.

Review

- **Confidence intervals** are measures of uncertainty around effect estimates. Interpretation of the frequentist 95% confidence interval: If we repeat the experiments many times, 95% of the constructed confidence intervals contains the true population.
- Interpretation of the Bayesian 95% confidence interval (which is known as **credible interval**): there is a 95% probability that the true parameter would lie within the interval, given the observed data.

Review

Bayes' Theorem

$$p(\theta|D) = \frac{P(D|\theta)p(\theta)}{p(D)}$$

- D : observed data
- θ : a generic symbol for the parameters in a particular model
- $p(\theta)$: **prior** belief of parameters before we collected any data
- $p(\theta|D)$: **posterior** belief of parameters after inferring data
- $P(D|\theta)$: likelihood of the data in view of the parameters
- $p(D)$: the **marginal likelihood**

Review

If we view the quantities as functions of θ , we can disregard the normalization constant $p(y)$.

$$\underbrace{p(\theta|D)}_{\text{posterior}} \propto \underbrace{p(D|\theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}$$

The conjugate prior distribution

is the family of probability distributions such that, when combined with the likelihood function, the posterior distribution falls in the same family.

Review

- After being trained, a model can be used to compute predictions ¹.
- Since θ is a random variable, the prediction connects to $p(\theta|\mathbf{y})$ via the marginalisation.

$$p(\mathbf{y}_*|\mathbf{y}) = \int p(\mathbf{y}_*|\theta)p(\theta|\mathbf{y})d\theta$$

¹Again this is a matter of computing a distribution $p(\mathbf{y}_*|\mathbf{y})$ (rather than a point prediction) for a test input $\mathbf{x}_*$

Section 2

Bayesian Linear Regression

Bayesian Linear Regression

Recall the SLR model

Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- Y_i : observed random variable
- β_0, β_1 : unknown deterministic parameter
- ε_i : unknown random variable

Bayesian Linear Regression

Two equivalent ways of expressing the linear regression model:

1 $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$

2 $p(Y_i | \beta_0, \beta_1) = \mathcal{N}(Y_i; \beta_0 + \beta_1 X_i, \sigma^2)$

The likelihood $p(\mathbf{y} | \beta_0, \beta_1)$ is given by

$$p(\mathbf{y} | \beta_0, \beta_1) = \prod_{i=1}^n p(y_i | \beta_0, \beta_1) = \prod_{i=1}^n \mathcal{N}(y_i; \beta_0 + \beta_1 x_i, \sigma^2)$$

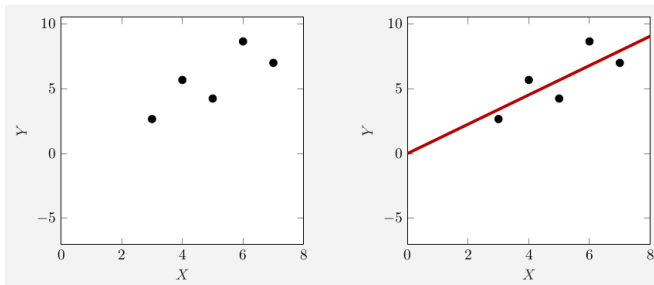
The solution is found by maximizing the likelihood

$$\hat{\beta}_0, \hat{\beta}_1 = \arg \max_{\beta_0, \beta_1} p(\mathbf{y} | \beta_0, \beta_1)$$

Bayesian Linear Regression

Example:

- Consider some training data
 $D = \{(3, 2.7), (4, 5.7), (5, 5.7), (6, 4.2), (7, 7.0)\}$, the black dots in the left panel below.
- We use a maximum likelihood linear regression model with no intercept term, i.e., $Y_i = \beta_1 X_i + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, 1)$.
- Least squares immediately gives $\hat{\beta}_1 = 1.13$, and $\hat{\beta}_1 X$ is plotted as a red line in the right panel.



Bayesian Linear Regression

Now we introduce a prior over the parameter β_0, β_1

Bayesian Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

$$\beta_0 \sim p(\beta_0), \beta_1 \sim p(\beta_1)$$

- Y_i - observed random variable
- β_0, β_1 - **random** parameters with prior distributions (different from SLR)
- ε_i - unknown random variable

Bayesian Linear Regression

For a scalar variable x , the Gaussian distribution can be written on the form

$$\mathcal{N}(x; \mu, \sigma^2) = \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}}}_Z e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ is the mean(expected value of the distribution)
- σ is the standard deviation
- σ^2 is the variance
- Z is a normalization constant

Bayesian Linear Regression

In Bayesian linear regression, we use a multivariate Gaussian distribution as prior over the parameters: $\boldsymbol{\beta} = (\beta_0, \beta_1)^T$

$$p(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\beta}; \mathbf{m}_0, \boldsymbol{\Sigma}_0)$$

Where: $\mathbf{m}_0$: prior mean vector

$\boldsymbol{\Sigma}_0$: prior covariance matrix

Bayesian Linear Regression

For a D -dimensional vector $\mathbf{x}$, the multivariate Gaussian distribution can be written on the form

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

- $\boldsymbol{\mu}$ is the mean vector
- $\boldsymbol{\Sigma}$ is the covariance matrix
- $Z = (2\pi)^{D/2} |\boldsymbol{\Sigma}|^{1/2}$ is the normalization constant

Bayesian Linear Regression

The posterior distribution is:

$$p(\boldsymbol{\beta}|\mathbf{y}) \propto p(\mathbf{y}|\boldsymbol{\beta})p(\boldsymbol{\beta})$$

where the proportionality ($\propto$ sign) is with respect to $\boldsymbol{\beta}$, we omit the $p(\mathbf{y})$, which does not depend on $\boldsymbol{\beta}$.

The likelihood $p(\mathbf{y}|\boldsymbol{\beta})$ given by

$$p(\mathbf{y}|\boldsymbol{\beta}) = \prod_{i=1}^n \mathcal{N}(y_i | \boldsymbol{\beta}^T \mathbf{x}_i, \sigma^2)$$

Equivalently, we can use the compact matrix notation,

$$p(\mathbf{y}|\boldsymbol{\beta}) = \mathcal{N}(\mathbf{y} | \mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$$

Bayesian Linear Regression

The posterior distribution is:

$$p(\boldsymbol{\beta}|\mathbf{y}) \propto p(\mathbf{y}|\boldsymbol{\beta})p(\boldsymbol{\beta})$$

The likelihood of data given the parameter:

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right)$$

The prior distribution:

$$p(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\beta}; \mathbf{m}_0, \boldsymbol{\Sigma}_0) = \frac{1}{(2\pi)^{p/2}|\boldsymbol{\Sigma}_0|^{1/2}} \exp\left(-\frac{1}{2}(\boldsymbol{\beta} - \mathbf{m}_0)^T \boldsymbol{\Sigma}_0^{-1}(\boldsymbol{\beta} - \mathbf{m}_0)\right)$$

Bayesian Linear Regression

The posterior distribution is given by Bayes' theorem:

$$p(\boldsymbol{\beta}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{X}, \boldsymbol{\beta}, \sigma^2) p(\boldsymbol{\beta})$$

Substituting the likelihood and prior:

$$p(\boldsymbol{\beta}|\mathbf{y}) \propto \exp \left(-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) - \frac{1}{2} (\boldsymbol{\beta} - \mathbf{m}_0)^T \boldsymbol{\Sigma}_0^{-1} (\boldsymbol{\beta} - \mathbf{m}_0) \right)$$

The posterior is also Gaussian:

$$p(\boldsymbol{\beta}|\mathbf{y}) = \mathcal{N}(\boldsymbol{\beta}; \mathbf{m}_n, \boldsymbol{\Sigma}_n)$$

Where:

$$\begin{aligned} \boldsymbol{\Sigma}_n &= \left(\boldsymbol{\Sigma}_0^{-1} + \frac{1}{\sigma^2} \mathbf{X}^T \mathbf{X} \right)^{-1} \\ \mathbf{m}_n &= \boldsymbol{\Sigma}_n \left(\boldsymbol{\Sigma}_0^{-1} \mathbf{m}_0 + \frac{1}{\sigma^2} \mathbf{X}^T \mathbf{y} \right) \end{aligned}$$

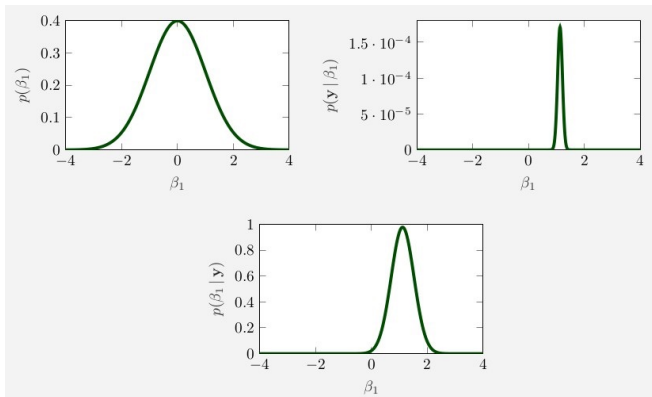
Bayesian Linear Regression

Consider the same data as in Example.

- Now we decide to do probabilistic linear regression with the model $Y = \beta_1 X$.
- For this, we also need to make a prior assumption about β_1 , which we decide to be $p(\beta_1) = N(\beta_1|0, 1)$, shown to the left below.
- The posterior, which we seek, is found using Bayes theorem, or simpler $p(\beta_1|Y) \propto p(\beta_1)p(Y|\beta_1)$. We have therefore plotted $p(Y|\beta_1)$ to the right below (note that this is not a probability density function in β_1).

Bayesian Linear Regression

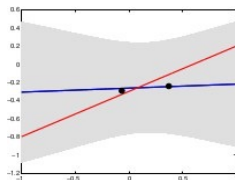
Inserting our prior and the data y into the expressions, we can see that the posterior distribution $p(\beta_1|y)$ is a Gaussian distribution with variance $\Sigma_N = 0.17$, mean $\mu_n = 1.12$, i.e., $p(\beta_1|y) = N(\beta_1|1.12, 0.17)$,



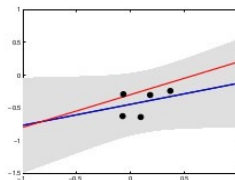
Section 3

Predictive Distribution

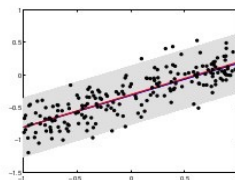
Predictive distribution



$N = 2$ observations



$N = 5$ observations



$N = 200$ observations

Gray shaded area: One standard deviation of the predictive distribution as function of x_* .

- Blue line: Mean of predictive distribution
- Black circles: Observations
- Red line: true model

Predictive distribution

Example: predict bodyfat using waist circumference (Abdomen)

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad i = 1, \dots, 252.$$

```
# Frequentist OLS linear regression
bodyfat.lm = lm(Bodyfat ~ Abdomen, data = bodyfat)
summary(bodyfat.lm)

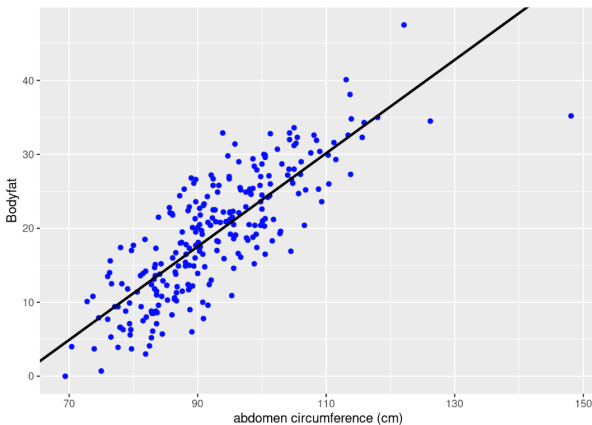
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -39.28018    2.66034  -14.77  <2e-16 ***
## Abdomen      0.63130     0.02855   22.11  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

Predictive Formula: $\widehat{\text{Bodyfat}} = -39.28 + 0.63 \times \text{Abdomen}.$

Predictive distribution

Example: predict bodyfat using waist circumference (Abdomen)

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad i = 1, \dots, 252.$$



Predictive distribution

Example: predict bodyfat using waist circumference (Abdomen)

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad i = 1, \dots, 252.$$

$$Y_i \mid x_i, \alpha, \beta, \sigma^2 \sim \text{Normal}(\alpha + \beta x_i, \sigma^2), \quad i = 1, \dots, n.$$

```
> bodyfat.bayes <- brm(  
+   formula = Bodyfat ~ Abdomen,  
+   data = bodyfat,  
+   family = gaussian(),  
+   prior = c(  
+     prior(normal(0, 100), class = Intercept),  
+     prior(normal(0, 100), class = b)  
+   ),  
+   chains = 2, iter = 2000  
+ )
```

Predictive distribution

Example: predict bodyfat using waist circumference (Abdomen)

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad i = 1, \dots, 252.$$

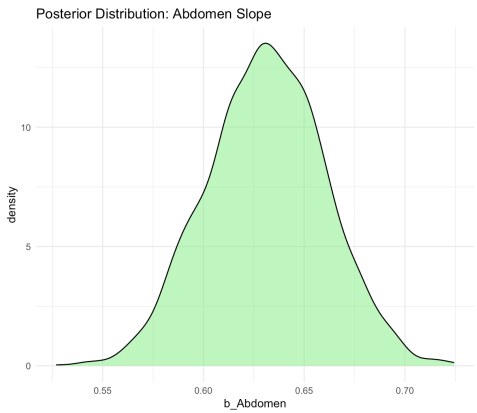
$$Y_i | x_i, \alpha, \beta, \sigma^2 \sim \text{Normal}(\alpha + \beta x_i, \sigma^2), \quad i = 1, \dots, n.$$

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	-39.27	2.72	-44.55	-33.95
Abdomen	0.63	0.03	0.57	0.69

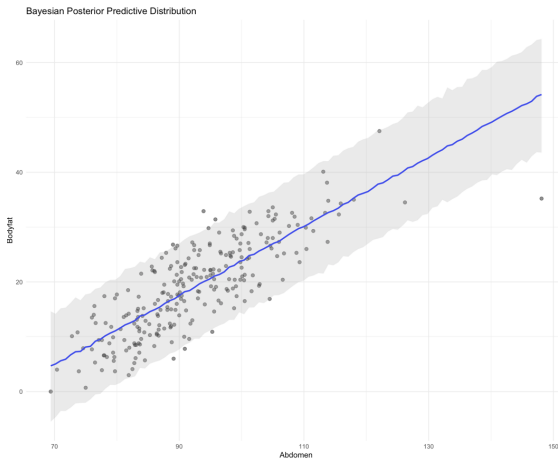
Predictive distribution

Example: predict bodyfat using waist circumference (Abdomen)



Predictive distribution

Example: predict bodyfat using waist circumference (Abdomen)



Predictive distribution

Example: predict bodyfat using waist circumference (Abdomen)

Informative prior:

```
> bodyfat.bayes.informative <- brm(  
+   formula = Bodyfat ~ Abdomen,  
+   data = bodyfat,  
+   family = gaussian(),  
+  
+   prior = c(  
+     prior(normal(0, 100), class = Intercept),  
+     prior(normal(0, 0.01), class = b)  
+   ),  
+  
+   chains = 2, iter = 2000  
+ )
```


Predictive distribution

Example: predict bodyfat using waist circumference (Abdomen)

Informative prior:

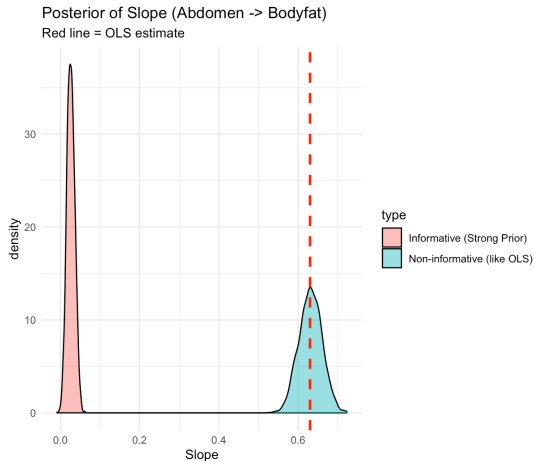
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	16.70	1.07	14.62	18.69
Abdomen	0.03	0.01	0.01	0.05

Predictive distribution

Example: predict bodyfat using waist circumference (Abdomen)

Informative prior:



Section 4

Regularization

Regularization

Recall:

- **Regularization** allows complex models to be trained on data sets of limited size without severe over-fitting, essentially by limiting the effective model complexity.

- **Ridge Regularization:**

$$SSE + \lambda \Sigma \beta^2$$

SSE: loss, sum of squared errors, measures fits to the data

$\lambda \Sigma \beta^2$: penalty, shrinkage term. $\lambda \geq 0$ controls strength, $\Sigma \beta^2$ is sum of squared coefficients

- balance fitting the data (SSE) with keeping coefficients small (Penalty).

Regularization

Bayesian Perspective: Prior as Regularization

$$\text{maximum: } p(\boldsymbol{\beta}|\mathbf{y}) \propto p(\mathbf{y}|\boldsymbol{\beta})p(\boldsymbol{\beta})$$

$$\log p(\boldsymbol{\beta}|\mathbf{y}) \propto \log p(\mathbf{y}|\boldsymbol{\beta}) + \log p(\boldsymbol{\beta})$$

Connection to Ridge Regression:

Bayesian log-posterior Term	Equivalent Ridge Regression Term
$\log p(\mathbf{y} \boldsymbol{\beta})$	$-\text{SSE}$ (fits the data)
$\log p(\boldsymbol{\beta})$	$-\lambda \boldsymbol{\Sigma} \boldsymbol{\beta}^2$ (penalty, regularization)

Regularization

Bayesian Perspective: Prior as Regularization

The prior term $p(\boldsymbol{\beta})$ in Bayesian regression plays the **exact same role** as the penalty term $\lambda \boldsymbol{\Sigma} \boldsymbol{\beta}^2$ in Ridge regression.

Gaussian Prior: $\boldsymbol{\beta} \sim \mathcal{N}(0, \sigma_{\boldsymbol{\beta}}^2 \mathbf{I})$

- The prior distribution $p(\boldsymbol{\beta}) \sim \mathcal{N}(0, \sigma_{\boldsymbol{\beta}}^2 \mathbf{I})$ represents our belief that coefficients $\boldsymbol{\beta}$ should be close to 0.
- Small $\sigma_{\boldsymbol{\beta}}^2$ means a stronger prior belief (coefficients are near 0), which leads to **stronger regularization and more shrinkage towards 0**.

Regularization

Bayesian Perspective: Prior as Regularization

The prior term $p(\boldsymbol{\beta})$ in Bayesian regression plays the **exact same role** as the penalty term $\lambda \boldsymbol{\Sigma} \boldsymbol{\beta}^2$ in Ridge regression.

Gaussian Prior: $\boldsymbol{\beta} \sim \mathcal{N}(0, \sigma_{\boldsymbol{\beta}}^2 \mathbf{I})$

- Large $\sigma_{\boldsymbol{\beta}}^2$ means a weaker prior, so the data dominates the posterior (similar to OLS).

Section 5

Summary

Summary

- Review
- Bayesian Linear Regression
- Predictive Distribution
- Regularization

Thank You !
Q & A